

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING AND TECHNOLOGY, KATTANKULATHUR CAMPUS
DEPARTMENT OF COMPUTING TECHNOLOGIES



Course Code	21CSC304J			L T P C	2 0 2 3
Course Title	Compiler Design			Faculty Name	Dr. M S Nagar
Semester	VI			Mail ID	mituln@srmist.edu.in
Pre-requisite / Co-Requisite	Nil			Slot & Class Hours:	Batch 1: DO3: 8.00-9.40 Batch 2: DO3: 12.30-2.15
Lecture Hall #	TP2- CLS1006	Laboratory Location	UB1213		

Organisation of the course							
Classes will be predominantly lectures, interlaced with invited guests, tutorials. There will be opportunities for group discussions, group tasks, in-class assignments.							
Course Outcome(s)							
1	Acquire knowledge of Lexical Analyzer from a specification of a language's lexical rules						
2	Apply different parsing algorithms to develop the parsers for a given grammar						
3	Gain knowledge to translate a system into various intermediate codes						
4	Analyze the methods of implementing a Code Generator for compilers						
5	Design the methods of developing a Code Optimizer						
Learning Materials							
T1. Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Compilers: Principles, Techniques and Tools, Second Edition, Pearson Education, 2011. T2. S. Godfrey Winster, S. Aruna Devi, R.Sujatha,”Compiler Design”, Yesdee Publishing Pvt.Ltd, 2016. T3. K .Muneeswaran,”CompilerDesign”, Oxford Higher Education, Fourth Edition, 2015. T4. David Galles, “Modern Compiler Design”, Pearson Education, Reprint 2012. T5. Raghavan V., “Principles of CompilerDesign”, Tata McGraw Hill Education Pvt. Ltd.,2010.							
Course Topics and Schedule							
Unit / Module No.	Topic Name	No. of Hours	Method of delivery (Multi-Select)	Assignment(s) / Activities	Learning materials to be referred	CO	Related PO

1	Compilers-Phases of Compiler	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Phases of Compiler	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Cousins of the Compiler, Grouping of Phases, Compiler construction tools	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Lexical Analysis	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Role of Lexical Analyze	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Role of Lexical Analyze	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Input Buffering	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Specifiction of Tokens, LEX	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Conversion of Reg Exp to DFA by Direct Method	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
1	Example for Conversion of Reg Exp to DFA by Direct Method	1	BB/Any other pedagogy		T1, T2	CO1	PO1, PO2, PO5
2	Role of Parser, Error Handling	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
2	Left Recursion- Left Factoring	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
2	Introduction to Top Down Parsing - Recursive Descent Parser	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
2	Computation of FIRST and FOLLOW	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
2	Constructing predictive parsing table	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
2	Predictive Parsers LL(1)	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
3	Bottom Up Parsing-Reductions-Handle Pruning-LEADING, TRAILING	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
3	Construction of SLR Parsing Table -Problems related to SLR	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
3	Construction of Canonical LR(1)- Problems related to CLR	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5

3	LALR Parser — Problems related to LALR-YACC	1	BB/Any other pedagogy		T1, T2	CO2	PO1, PO2, PO5
4	Intermediate Code Generation- prefix – postfix notation	1	BB/Any other pedagogy		T1, T2	CO3	PO1, PO2, PO5
4	Quadruple - triple - indirect triples Representation - Syntax tree- Evaluation of expression	1	BB/Any other pedagogy		T1, T2	CO3	PO1, PO2, PO5
4	Synthesized attributes – Inherited attributes	1	BB/Any other pedagogy		T1, T2	CO3	PO1, PO2, PO5
4	Intermediate languages – Declarations- Assignment Statements- Boolean Expressions- Case Statements	1	BB/Any other pedagogy		T1, T2	CO4	PO1, PO2, PO4
4	Back patching – Procedure calls- Code Generation- Issues in the design of code generator	1	BB/Any other pedagogy	Activity	T1, T2	CO4	PO1, PO2, PO4
5	The target machine – Runtime Storage management- A simple Code generator-	1	BB/Any other pedagogy		T1, T2	CO4	PO1, PO2, PO4
5	Code Generation Algorithm- Register and Address Descriptors	1	BB/Any other pedagogy		T1, T2	CO5	PO1, PO2, PO3, PO4, PO5
5	Code optimization -Principal Sources of Optimization- Function Preserving Transformation	1	BB/Any other pedagogy		T1, T2	CO5	PO1, PO2, PO3, PO4, PO5
5	Loop Optimization- Peephole optimization	1	BB/Any other pedagogy	Activity	T1, T2	CO5	PO1, PO2, PO3, PO4, PO5
Grading Criteria							
Type of assessments	Deadline	Weightage	Due Date (if applicable)	Component			
FJ-I	Week-7	15%	26.02.26	Written Test - 1 and Completion of Lab Experiments and Viva			
FJ-II	Week-12	15%	08.04.26	Written Test - 2 and Completion of Lab Experiments and Viva			
FJ-III	Week-15	15%	25 th April 26 – 5 th May 2026	Worksheet Assessment and Activity - Scenario based Problems Chart Presentation			
LLJ-I	Week-9	7%	7 th March 2026 – 15 th March 2026	Assessment of LAB Mid Term (Expr 4-8)			

LLJ-II	Week-15	8%	27 th April 2026 – 01 st May 2026	Assessment of Lab FINAL test (Expr 9-12)			
Continuous Assessment	Total	60%					
End Semester Examination		40%					
	Total	100%					
Make-up Policy							
<i>Assignments / Term Project</i>							
Proper academic performance depends on students doing their work not only well, but on time. Accordingly, assignments for this course must be received on the due date specified for the assignment. No Make-up will be available for assignments or term project. Late submissions will be evaluated at 25% less weight for that component for a delay of up to 24 hours after which no submissions will be accepted.							
Test							
Make-up test is usually not permitted, but in case of very genuine reasons as decided by the course faculty, course coordinator/audit professor special prior Permission is usually required to get a make-up test. It cannot be taken as a right to claim the test marks.							
Academic Integrity							
<i>Plagiarism or cheating will result in a grade of zero for the assignment or exam. Severe cases may result in course failure.</i>							
Contact Information							
Feel free to reach out via email for any questions regarding the course material. For urgent matters, you can visit the faculty room during office hours.							